

NOAA Northeast Fisheries Science Center: New England Offshore – Aerial Surveys 2024-2027

Type Enterprise Geodatabase Feature Class

Tags NOAA, NEFSC, Aerial Surveys, Marine Mammals, Whales, New England, Offshore

Summary

This layer depicts transects from the NOAA-Northeast Fishery Science Center’s aerial surveys that are conducted from 2024-2027 in the New England Offshore area. This layer contains three separate track lines that make up NEFSC’s survey effort for the entire NE Offshore area. These lines show only the planned routes for the aerial surveys and are displayed for planning and research coordination purposes. Realized routes can be found on <https://whalemap.org/>.

Description

This layer depicts planned transects from NOAA-Northeast Fishery Science Center’s aerial surveys conducted from 2024-2027 in the NE Offshore Area. For more on methodology and project details visit: <https://www.fisheries.noaa.gov/new-england-mid-atlantic/marine-mammal-protection/whale-and-dolphin-research-northeast#north-atlantic-right-whale-sighting-surveys> ; and the RWSC database project page: <https://database.rwsc.org/details?recordId=rec5rsRHi5cVbXmAk> . Realized routes can be found on <https://whalemap.org/#map>.

Credits

Tim Cole - NOAA tim.cole@noaa.gov

Use limitations

There are no access and use limitations for this item.

Extent

West -73.300000 East -69.033333
North 41.591667 South 40.000000

Scale Range

Maximum (zoomed in) 1:5,000
Minimum (zoomed out) 1:150,000,000

Topics and Keywords ▶

Content type ⇌ Downloadable Data

Citation ▶

Title NOAA Northeast Fisheries Science Center: New England Offshore – Aerial Surveys 2024-2027
Creation date 2025-05-02 00:00:00
Publication date 2025-08-19 00:00:00

Presentation formats ⇌ digital map

Resource Details ▶

Dataset languages ⇌ English (UNITED STATES)
Dataset character set utf8 - 8 bit UCS Transfer Format

Status on-going
Spatial representation type ⇌ vector

Supplemental information
This layer depicts transects from NOAA-Northeast Fishery Science Center’s aerial surveys conducted from 2024-2027 in the NE Offshore Area. The NEFSC’s survey of the NE Offshore area is split into four separate effort areas with individual track lines, although sometimes multiple areas are completed within a single outing. The areas include South of Martha’s Vineyard and Nantucket, Southeast of Nantucket, Block Canyon, and South of Long Island. Data collection focuses on six large whale species - blue, fin, sei, humpback, sperm, and North Atlantic right whales - but also includes opportunistic sightings of sea turtles and other marine animals. The objectives are: (1) to determine each whale species’ distribution and estimate density; (2) to determine each whale species’ monthly and seasonal occurrence; (3) to characterize the interannual variability of the whales’ distributions and densities; and (4) to record behavioral data of sighted whales as much as possible. Summaries will be publicly available as surveys progress.

These surveys are a continuation of the NEFSC Right Whale Aerial Survey program that has been running since 1998. The program began with patrols of Great South Channel and transitioned in 2002 to systematic line surveys of the Northeastern United States Exclusive Economic Zone from the Gulf of Maine to South of Martha’s Vineyard and Nantucket. In 2006, the program transitioned to box surveys in the Gulf of Maine and Southern New England in order to increase the number of right whales identified per flight hour. In 2018, box survey efforts began south of New England. These box surveys continued through 2023, with special attention to right whale aggregations within the boxes after 2016. In 2024 the surveys were extended to the current coverage and converted to the line surveys seen in the layer.

The survey team used a de Havilland Twin Otter aircraft flown at 750 to 1500 feet at 100-110 knots. The routes South of Nantucket and Martha’s Vineyard, southeast of Nantucket, Block Canyon, and South of Long Island are flown 6 times a year (every 1-2 months).

These lines show only the planned routes for the aerial surveys and are displayed for planning and research coordination purposes. Realized routes can be found on <https://whalemap.org/>.

For more information visit the project site: <https://www.fisheries.noaa.gov/new-england-mid-atlantic/marine-mammal-protection/whale-and-dolphin-research-northeast#north-atlantic-right-whale-sighting-surveys>; and the RWSC database project page: <https://database.rwsc.org/details?recordId=rec5rsRHi5cVbXmAk>

Processing environment ⇌ Microsoft Windows 10 Version 10.0 (Build 26200) ; Esri ArcGIS 13.7.0.1901

Credits
Tim Cole - NOAA tim.cole@noaa.gov

ArcGIS item properties

Name ⇌ rpt.rpt.NEFSC_NEOffshore_AerialSurveys
Size ⇌ 0.000
Location ⇌ Server=rwsc-db-pg15.env.duke.edu; Service=sde:postgresql:rwsc-db-pg15.env.duke.edu; Database=rpt; User=rpt; Version=sde.DEFAULT
Access protocol ⇌ ArcSDE Connection

Extents ►

Extent

Geographic extent

Bounding rectangle

Extent type

Extent used for searching

West longitude ⇌ -73.300000

East longitude ⇌ -69.033333

North latitude ⇌ 41.591667

South latitude ⇌ 40.000000

Vertical extent

Minimum value ⇌ 0.000000

Maximum value ⇌ 0.000000

Extent in the item's coordinate system

westBL ⇌ -73.300000

eastBL ⇌ -69.033333

southBL ⇌ 40.000000

northBL ⇌ 41.591667

exTypeCode ⇌ Yes

Resource Maintenance ►

Resource maintenance

Update frequency ⇌ as needed

Spatial Reference ►

ArcGIS coordinate system

Type ⇌ Geographic

Geographic coordinate reference ⇌ GCS_WGS_1984

Coordinate reference details ⇌

GeographicCoordinateSystem

WKID ⇌ 4326

XOrigin ⇌ -400

YOrigin ⇌ -400

XYScale ⇌ 1111948722.2222219

ZOrigin ⇌ -100000

ZScale ⇌ 10000

MOrigin ⇌ -100000

MScale ⇌ 10000

XYTolerance ⇌ 8.983152841195215e-09

ZTolerance ⇌ 0.001

MTolerance ⇌ 0.001

HighPrecision ⇌ true

LeftLongitude ⇌ -180

LatestWKID ⇌ 4326

WKT

GEOGCS["GCS_WGS_1984",DATUM["D_WGS_1984",SPHEROID["WGS_1984",6378137.0,298.257223563]],PRIMEM["Greenwich",0.0],UNIT["Degree",0.0174532925199433],AUTHORIT

Reference system identifier

Value ⇌ 4326

Codespace ⇌ EPSG

Version ⇌ 6.2(3.0.1)

Spatial Data Properties ►

Vector ►

Level of topology for this dataset ⇌ geometry only

Geometric objects

Feature class name ⇌ rpt.rpt.NEFSC_NEOffshore_AerialSurveys

Object type ⇌ composite

Object count ⇌ 4

ArcGIS Feature Class Properties ►

Feature class name ⇌ rpt.rpt.NEFSC_NEOffshore_AerialSurveys

Feature type ⇌ Simple

Geometry type ⇌ Polyline

Has topology ⇌ FALSE

Feature count ⇌ 4

Spatial index ⇌ TRUE

Linear referencing ⇌ TRUE

Distribution ►

Distribution format
Name ⇔ Enterprise Geodatabase Feature Class

Transfer options
Transfer size ⇔ 0.000

Fields ►

Details for object rpt.rpt.NEFSN_Eoffshore_AerialSurveys ►
Type ⇔ Feature Class
Row count ⇔ 4

Field objectid ►
Alias ⇔ objectid
Data type ⇔ OID
Width ⇔ 4
Precision ⇔ 10
Scale ⇔ 0

Field description ⇔
Internal feature number.

Description source ⇔
Esri

Description of values ⇔
Sequential unique whole numbers that are automatically generated.

Field Shape ►
Alias ⇔ shape
Data type ⇔ Geometry
Width ⇔ 8
Precision ⇔ 0
Scale ⇔ 0

Field description ⇔
Feature geometry.

Description source ⇔
Esri

Description of values ⇔
Coordinates defining the features.

Field OPERATOR ►
Alias ⇔ OPERATOR
Data type ⇔ String
Width ⇔ 254
Precision ⇔ 0
Scale ⇔ 0

Field POC_EMAIL ►
Alias ⇔ POC_EMAIL
Data type ⇔ String
Width ⇔ 254
Precision ⇔ 0
Scale ⇔ 0

Field PROJECT_NA ►
Alias ⇔ PROJECT_NA
Data type ⇔ String
Width ⇔ 254
Precision ⇔ 0
Scale ⇔ 0

Field PATH_NAME ►
Alias ⇔ PATH_NAME
Data type ⇔ String
Width ⇔ 254
Precision ⇔ 0
Scale ⇔ 0

Field START_YEAR ▶
Alias ↔ START_YEAR
Data type ↔ Integer
Width ↔ 4
Precision ↔ 10
Scale ↔ 0

Field END_YEAR ▶
Alias ↔ END_YEAR
Data type ↔ Integer
Width ↔ 4
Precision ↔ 10
Scale ↔ 0

Field START_DATE ▶
Alias ↔ START_DATE
Data type ↔ String
Width ↔ 254
Precision ↔ 0
Scale ↔ 0

Field END_DATE ▶
Alias ↔ END_DATE
Data type ↔ String
Width ↔ 254
Precision ↔ 0
Scale ↔ 0

Field FREQUENCY ▶
Alias ↔ FREQUENCY
Data type ↔ String
Width ↔ 254
Precision ↔ 0
Scale ↔ 0

Field PROJECT_LI ▶
Alias ↔ PROJECT_LI
Data type ↔ String
Width ↔ 254
Precision ↔ 0
Scale ↔ 0

Field DATE_SUBMI ▶
Alias ↔ DATE_SUBMI
Data type ↔ Date
Width ↔ 8
Precision ↔ 0
Scale ↔ 0

Field DATE_ADDED ▶
Alias ↔ DATE_ADDED
Data type ↔ Date
Width ↔ 8
Precision ↔ 0
Scale ↔ 0

Field Id ▶
Alias ↔ Id
Data type ↔ Integer
Width ↔ 4
Precision ↔ 10
Scale ↔ 0

Field Name ▶

Alias ↔ Name
Data type ↔ String
Width ↔ 254
Precision ↔ 0
Scale ↔ 0

Field FolderPath ▶

Alias ↔ FolderPath
Data type ↔ String
Width ↔ 254
Precision ↔ 0
Scale ↔ 0

Field SymbolID ▶

Alias ↔ SymbolID
Data type ↔ Integer
Width ↔ 4
Precision ↔ 10
Scale ↔ 0

Field AltMode ▶

Alias ↔ AltMode
Data type ↔ Integer
Width ↔ 4
Precision ↔ 10
Scale ↔ 0

Field Base ▶

Alias ↔ Base
Data type ↔ Double
Width ↔ 8
Precision ↔ 38
Scale ↔ 8

Field Clamped ▶

Alias ↔ Clamped
Data type ↔ Integer
Width ↔ 4
Precision ↔ 10
Scale ↔ 0

Field Extruded ▶

Alias ↔ Extruded
Data type ↔ Integer
Width ↔ 4
Precision ↔ 10
Scale ↔ 0

Field Snippet ▶

Alias ↔ Snippet
Data type ↔ String
Width ↔ 254
Precision ↔ 0
Scale ↔ 0

Field PopupInfo ▶

Alias ↔ PopupInfo
Data type ↔ String
Width ↔ 254
Precision ↔ 0
Scale ↔ 0

Field Shape_Leng ▶
Alias ↔ Shape_Leng
Data type ↔ Double
Width ↔ 8
Precision ↔ 38
Scale ↔ 8

Field st_length(shape) ▶
Alias ↔ st_length(shape)
Data type ↔ Double
Width ↔ 0
Precision ↔ 0
Scale ↔ 0

Metadata Details ▶

Metadata language ↔ English (UNITED STATES)
Metadata character set ↔ utf8 - 8 bit UCS Transfer Format

Scope of the data described by the metadata ↔ dataset
Scope name ↔ dataset

Last update ↔ 2026-08-24

ArcGIS metadata properties
Metadata format ArcGIS 1.0
Standard or profile used to edit metadata FGDC

Created in ArcGIS for the item 2024-01-24 12:31:40
Last modified in ArcGIS for the item 2026-08-24 10:29:09

Automatic updates
Have been performed Yes
Last update 2026-08-24 10:29:09

Item location history
Item copied or moved 2024-01-24 12:31:40
From C:\Users\jozog\OneDrive - HDR, Inc\GIS\HDR_NARW_Aerial_Tracklines
To \\DZ7YXT3\C\$\Users\jozog\OneDrive - HDR, Inc\GIS\Aerial_for_Jackie\HDR_NARW_Aerial_Tracklines